Technology in Society 82 (2025) 102934



Contents lists available at ScienceDirect

Technology in Society

journal homepage: www.elsevier.com/locate/techsoc





Assessing the role of human factor in digital transformation projects: A systematic literature review and research agenda

Maria Anna Lykourentzou [a], Nikolaos Apostolopoulos [a], Marina Dabić [b,c,d,*], Panagiotis Liargovas [a], Metka Tekavčič [d]

[a] Department of Management Science and Technology, University of the Peloponnese, 22100, Tripoli, Greece
[b] University of Zagreb Faculty of Economics and Business, 10000, Zagreb, Croatia
[c] University of Dubrovnik, 20000, Dubrovnik, Croatia
[d] University of Ljubljana School of Economics and Business, 1000, Ljubljana, Slovenia

ARTICLE INFO

Keywords:
Digital transformation projects
Systematic literature review
Content analysis
Business processes
Business model
Operational management plan

ABSTRACT

Extensive research has been conducted on key corporate factors that have been impacted, and ultimately transformed, by digital transformation projects. These factors have been broadly analysed as distinct categories, the combination of which has led to a quad-factor model, which has already been introduced to applicators for the future successful delivery of digital transformation projects. Following a structured evaluation model of articles, this study presents a systematic literature review of 183 academic articles published between 2014 and 2023 to assess whether this quad-factor model provides a sufficiently robust framework for successful application of a digital transformation project, or whether the human factor needs to be introduced as the fifth independent factor supporting a project's results. Moreover, this study proposes a theoretical penta-factor model as a conceptual framework which presents human factor to have an equal contribution to the success of a digital transformation project. This will be helpful to future applicators in their efforts to design and deliver successful digital transformation projects with enhanced success and acceptance rates across the corporate ecosystem. Contributing to this highly relevant topic, the study summarises the ongoing trends in digital transformation projects and proposes promising avenues for future research.

1. Introduction

The extensive growth of technological advancements related to corporate development has driven organisations to invest in their digital modernisation. Corporate digital modernisation has been extensively analysed and documented, leading to the introduction of the term 'digital transformation', which describes the extensive technological changes that organisations must apply to enter the digital era.

According to Palagan (2025), 'Digital transformation (DT) serves as a catalyst for business model innovation by strategically implementing data-driven technologies that enhance decision-making, optimize processes, and create new avenues for organizational growth'. The early adopters of digital transformation were focused on improving the tangible corporate elements, such as business processes, business model, and organisational restructure while presenting substantial benefits for the companies. These elements have been extensively analysed in academic literature as independent contributing factors of a comprehensive and effective DT project, highlighting the significant modifications to traditional application models (Zaoui & Souissi, 2020). This systematic literature review aims to initially present the impact that a DT project has on key corporate factors, by sufficiently addressing the following question:

RQ1. *How are the four fundamental corporate factors (business processes, business model, organisational model, and corporate data) impacted by the DT projects of organisations?*

Digital transformation as a research field receives significant attention in both academic and practical applications. Recent studies have highlighted a new corporate component, the 'human factor', which appears to alter and impact the results of a DT project. To date, most academic publications were evaluating the core corporate factors excluding their connection to human attributes, such as employee

* Corresponding author. University of Zagreb Faculty of Economics and Business, 10000, Zagreb, Croatia.
E-mail addresses: lykourentzou@go.uop.gr (M.A. Lykourentzou), anikos@uop.gr (N. Apostolopoulos), mdabic@net.efzg.hr (M. Dabić), liargova@uop.gr (P. Liargovas), Metka.tekavcic@ef.uni-lj.si (M. Tekavčič).

https://doi.org/10.1016/j.techsoc.2025.102934
Received 7 July 2024; Received in revised form 22 April 2025; Accepted 29 April 2025
Available online 30 April 2025
0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

performance, corporate leadership, and customer loyalty (Williams & van Triest, 2023). Human factor was primarily considered a secondary element, as the primary target was the efficient technological implementation and maximisation of benefits. The assumption that the technological infrastructure combined with advanced software would be sufficient to achieve the desired results was extensively embraced. Employees were perceived as passive users of the new technologies, rather than active contributors to the digital transformation. However, as academic research indicates, this perception was followed by resistance to change, low adoption rates, a lack of enthusiasm and dissatisfaction with the new systems, highlighting the importance and involvement of the human factor (Strohmeier, 2020).

Considering the current analysis, this study aims to evaluate whether the human factor could be considered an independent corporate factor in future applications that should be assessed within a DT project. Although technology is the driving force underlying DT, its success is heavily dependent on the people involved – employees, leaders, customers, and stakeholders, and their acceptance of the new technological ecosystem. This study aims to contribute to this stream of literature analysis and review by addressing the following research question:

RQ2. *How is the human factor as one of the fundamental corporate factors impacted by DT projects?*

While addressing these questions, this study provides a comprehensive analysis on the matter, aiming to address the existing gap in the academic literature – particularly through emphasising the critical importance of the human factor. The shift identified in most recent applications and theoretical approaches has occurred as a result of organisations and DT applicators realising that the people of an organisation, corporate culture, and leadership adaptability are essential contributing elements for the successful application of DT projects (Gaß, Ortbach, Kretzer, Maedche, & Niehaves, 2015). This recognition has led to the development of human-centric approaches, focusing on employee contribution, digital literacy, and changes in the organisational management. The findings presented in this study offer both theoretical insights and practical guidance for organisations embarking on their DT journeys, showcasing to future applicators that only a balance between technological advances and human-centric approaches can support sustainable corporate success in the digital age.

2. Theoretical background

DT, as a process, is frequently adopted by organisations when they evolve and grow, driven by either internal or external factors. Vial (2019) defined DT as 'a process by which organisations use digital technologies to fundamentally change their operations, business models and customer experiences. This process involves the integration of digital tools and platforms across all levels of the organization, enabling new ways of creating value, enhancing operations efficiency and driving innovation'. Contrarily, DT projects have been defined as 'time-based, structured initiatives undertaken by organisations, to implement digital technologies with pre-defined goals, deliverables and expectations' (Kane et al., 2019).

Considering the two indicative definitions of the terms DT and DT projects, several distinctions can be underlined. DT is generally considered to be an on-going, organisation-wide conceptual process, whereas DT projects are expected to be finite, time-bound initiatives. Another important differentiation is that DT is routinely considered to be a long-term endeavour that seeks to shift the organisation towards integrating new technologies while fostering a digital culture. Conversely, DT projects have specific, pre-defined objectives and deliverables, while the size of the impact within the organisation varies depending on the management's expectations and vision.

This study uses DT projects as its basis. This is because they produce specific outcomes, and the results of the transformation are visible and documented during all the phases of the project.

2.1. Digital transformation: a multifunctional solution

DT projects aim to improve several corporate factors, such as increasing employee efficiency, company profitability (Ivančić et al., 2019), and customer satisfaction (Mergel et al., 2019). The first component is frequently achieved by digitising internal processes and procedures, automating manual tasks, creating automated workflows (Verhoef et al., 2021; Feroz et al., 2023), and hiring new talent having technological knowledge and experience (Zhang & Chen, 2024).

For most modern companies, their current technological infrastructure comprises multiple independent systems and services, which most frequently require human involvement for coordination and facilitation (Spaltini, Acerbi, Pinzone, Gusmeroli, & Taisch, 2022). Their limited capabilities restrict the companies' growth and evolution in a continuously evolving and demanding environment (Zhu et al., 2021). Therefore, updating all existing technological infrastructure with state-of-the-art energy efficient and multifunctional equipment (Pappas, Mikalef, Giannakos, Krogstie, & Lekakos, 2018) would benefit and support the current and future needs of the company.

Another attribute is increasing company profitability by enhancing the business model to support company growth and capture new markets and audiences (Zhu et al., 2021). Re-designing current sales' channels and developing new channels allows companies to reach out to larger and different audiences and create new revenue streams, while freeing themselves from the restrictions of physical location and time zones. Simultaneously, customer satisfaction can be enhanced by processing requests rapidly, where customers can receive notifications and status updates, and complete transactions remotely without physical contact or exchanging physical documents (Kraus et al., 2021).

DT projects tend to form part of a company's greater corporate strategic plan (Loonam, Eaves, Kumar, & Parry, 2018). As such transformative projects include risks and impact the current corporate operational model, a detailed analysis including all modifications should be produced and evaluated by company administrators prior to any implementations (Hashem et al., 2015). The project team overseeing this major transformation should possess an in-depth understanding of the corporate mechanism of functioning (i.e. processes, procedures, organisational management plan) (Matt et al., 2015), along with sound knowledge on several technological tools and methodologies (Zaki, 2019). Accordingly, such teams would be able to provide functional and sustainable solutions for their companies, targeting aspects such as employee productivity and efficiency, increased profits, and customer satisfaction (Tavoletti, Kazemargi, Cerruti, Grieco, & Appolloni, 2021).

2.2. Digital transformation during the COVID-19 pandemic

The COVID-19 pandemic caused an explosion of demand for DT projects in both public- and private-sector organisations (Datta & Nwankpa, 2021), as government regulations enforced worldwide almost eliminated physical commercial transactions among people and organisations. Consequently, the commerce ecosystem was compelled to transition from physical to digital transactions, either by activating new digital channels or enhancing and modernising those already in place (Kudyba, 2020).

Typically, a DT project would require extensive planning and calculation of the key components (i.e. time, cost, effort, change management) required for its successful implementation (Sehlin et al., 2019). During the COVID-19 pandemic, the demand for transitioning from physical to digital channels was extensive, and occasionally risked the quality of deliverables (Kempeneer & Heylen, 2023). Without following all the proper procedures and protocols, the systems in place were vulnerable to potential risks and failures in the future. Finally, for the companies compelled to invest in DT projects, their limited budgets led to delays in delivery or to the delivery of minimum viable products (MVPs) (Li, 2020).

3. Methodological approach

This literature review analysis predominantly aims to identify and analyse research articles focusing on corporate factors that are impacted during DT projects. Further, the validity of the current findings and the identification of outcomes and gaps for future research analysis are presented (Kitchenham, 2004; Gaß, Ortbach, Kretzer, Maedche, & Niehaves, 2015). The methodological approach followed is the Systematic Literature Review (SLR) as it proved to be the most suitable for our scope of research.

Systematic literature review (SLR) is widely recognised in academia as a robust and credible research methodology. In this study, SLR was selected as the most appropriate approach owing to the substantial number of academic articles analysed (Petticrew & Roberts, 2006). It provides a rigorous framework for identifying patterns, conceptual relationships, and research gaps across literature (Tranfield et al., 2003). This methodology fully supports the objective of this study—not only to cluster and thematically categorise the selected theoretical and empirical articles, but also to explore their synergies and conduct an impact analysis.

Existing literature has established that DT projects influence several core corporate factors. By applying SLR, this study aims to derive evidence-based conclusions regarding whether the human factor should be considered an independent and fundamental corporate dimension or whether it should be integrated within existing constructs (processes, organisational structure, data or leadership).

Furthermore, the SLR methodology offers a transparent and structured process for literature selection (see Fig. 1), including coding procedures, inclusion and exclusion criteria, and reduction techniques (Dabić et al., 2019). This process enables to identify a final set of academic sources that align with predefined standards of relevance, quality, and contribution. Ultimately, this review is expected to uncover specific thematic areas that warrant further investigation and contribute to addressing current research gaps in implementing DT projects (Okoli & Schabram, 2010).

The empirical and theoretical articles used as the basis for the current analysis were published between 2014 and 2023 – the selected research period – since Industry 4.0 began in earnest in and around 2013 (Abiodun et al., 2023). This period represents the most transformative era in terms of technological advancement and organisational change, marked by the widespread adoption of key technologies, such as automation, artificial intelligence (AI), big data, the internet of things (IoT), and cloud computing – all of which are critical drivers of DT (Grybauskas et al., 2022). Additionally, the 2014–2023 timeframe captures the global push for DT accelerated by such significant events as the COVID-19 pandemic, which further compelled organisations to rapidly digitise operations, embrace remote work, and offer an innovative customer experience through their digital channels (Vratimos, 2023). Therefore, this period provides a comprehensive view of how DT projects support evolution of companies, making it ideal for academic analysis.

3.1. Systematic review of academic literature

The systematic literature review method has been extensively used in previous studies owing to its ability to provide a guided, structured method of eliminating articles and journals, thereby yielding the most valid datasets for scholars (Kraus et al., 2021). The method used in this study is described below (see Fig. 2). The first step involved identifying keywords (Sutton et al., 2016), which were then ascribed to two categories: one associated with the 'digital transformation projects' perspective and the other related to the impacted corporate factors. The first group of keywords included such terms as 'digital transformation project' AND ['COVID-19' OR ('successful application' OR 'benefits' OR 'challenges' OR 'limitations' OR 'risks')]. The second group of keywords included such terms as 'digital transformation project' AND [('processes' OR 'business model') OR ('organisation' OR 'structure'), OR ('big data' OR 'data analytics'), OR ('customer experience' OR 'human factor' OR 'employee' OR 'manager' OR 'leadership')]. These terms emerged from the analysis of key research in various fields, including economics, technology, sociology, and workplace academic studies.

To ensure the completeness of the keywords, an additional evaluation was conducted (as detailed below). A preliminary search was conducted in Scopus, Web of Science (WoS), and SAGE databases, focusing on 'title, abstract, and keywords' to gather different datasets of papers. The Scopu, WoS, and SAGE searches yielded 1,663, 826, and 1237 papers, respectively. To verify the thoroughness of the evaluation, the titles, abstracts, and keywords were analysed using VOSviewer. The analysis confirmed that the selected keywords were sufficiently comprehensive, and no additional terms were required.

The next step involved the initial skimming phase, during which the set of collected papers was refined according to specific inclusion and exclusion criteria: source type (journals or reviews), language (English), publication period (2014–2023), and subject area (all relevant categories) (Teichert, 2019). Conference papers were excluded as they generally have a lower scientific impact and robustness level, while all subject categories were included to capture the application of DT projects. Thereafter, all the datasets were merged, with duplicate papers removed, leaving a dataset of 419 papers. In the second skimming phase, the titles, abstracts, and keywords of the remaining papers were reviewed in detail to determine their relevance to the research topic and objectives (Siddaway et al., 2019), creating a final dataset of 183 papers. Two independent peer reviewers evaluated the papers using a pre-established content assessment protocol to minimise bias and ensure validity.

For the data analysis, content analysis was selected as the most appropriate approach, considering its focus on the coding, categorisation, and systematic grouping of textual data (Acar & Puntoni, 2016). The methodology of this research involves the review of 183 academic articles, which are clustered and synthesised into structured thematic sections. This process aims to offer an overview of current research trends, the frequency of conceptual occurrences, and evolution of corporate factors in digital transformation (DT) projects (Elo & Kyngäs, 2008). Using this approach, the study seeks to determine whether traditionally acknowledged corporate factors remain relevant and assess the prominence and conceptual treatment of the human factor within existing literature.

Content analysis is particularly suited for analysing large volumes of sources, as it enables the systematic extraction of frequency-based data while maintaining transparency and replicability in the coding process

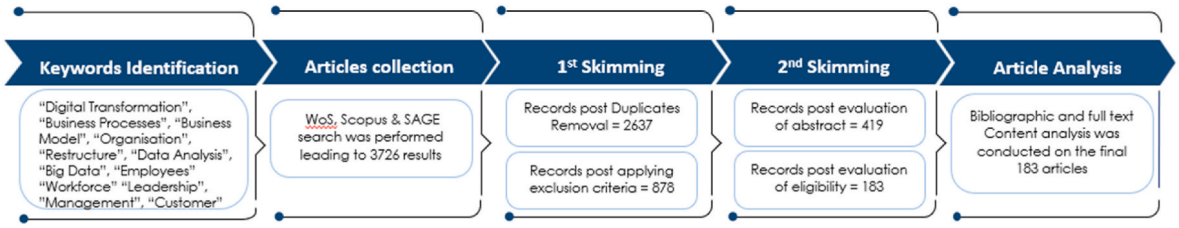


Fig. 1. Summary of systematic literature review methodology.

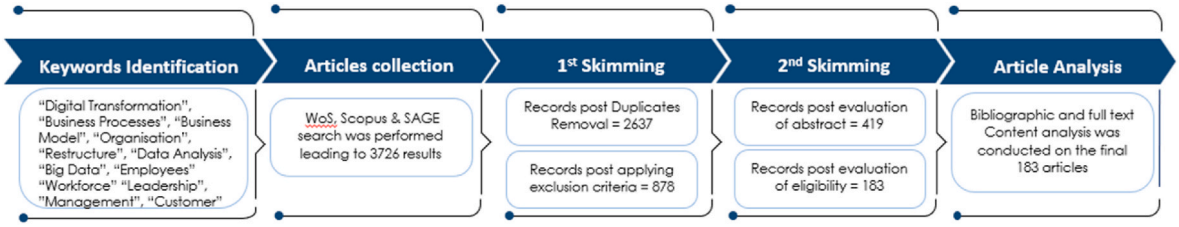


Fig. 2. Steps of Academic Literature Review analysis.

(Neuendorf, 2017). It complements the systematic literature review (SLR) framework by reducing researcher bias and emphasising methodological objectivity. Moreover, it allows identifying recurring concepts and gaps in literature, ensuring alignment with the research questions (Mayring, 2000). As the study aims to map and structure the current body of academic knowledge to support developing an updated theoretical framework for DT project implementation, content analysis is well aligned with this objective (Vaismoradi et al., 2013).

4. Results

4.1. Thematic analysis of academic literature

Studies on DT projects have thoroughly analysed all the corporate factors transformed during and post their implementation, such as the internal process, business model, organisational management plan, and customer experience (Doukidis et al., 2020). However, only a few studies have captured and analysed the human factor, despite it pertaining to all the above factors as a success or failure contributor to the application of a DT project. To capture the impact of the human factor on modifications and updates of corporate factors during a DT project (Berman, 2012; Neumann, Winkelhaus, Grosse, & Glock, 2021), we focused not only on the systemic changes, which included the technological advancements and processes, but also on the organisational changes that are required to occur simultaneously. Identifying the core transformed corporate factors of a DT project, as well as the means through which to guarantee success would create a holistic theoretical framework that could improve the application and success rates of future DT projects.

From the bibliographic analysis conducted, five key corporate factors emerged, which were used as the bases of the literature review. The procedures followed and the results from the used tools indicated a strong notion towards the extensive involvement of the human factor in DT projects. Therefore, the first four factors followed the quad-pillar model structure (Doukidis et al., 2020), whereas the fifth (i.e. the human factor) was included as our contribution to the literature. The final categories of clustered articles that constitute the penta-model are: articles focused on changes of internal organisational processes and procedures (DTCF1), articles with strategies for redesigning a corporate business model (DTCF2), articles including the impact of a DT project on the organisational management plan (DTCF3), articles focusing on the changes produced by data-driven decisions (DTCF4), and articles emphasising the involvement of people (employees, management, customers) in all the stages of DT projects (DTCF5).

Fig. 3 presents the DTCF corporate factors and their core subcategories, which were further analysed in the content analysis. Although several terms and concepts were identified, only those which were more frequently mentioned were included in the current paper's review.

4.2. Content analysis of academic literature

Following the bibliographic analysis, a detailed content analysis was executed on the final 183 articles, during which they were clustered based on their research topic (Higgins, 2019; Earley, 2014). Over 66 % of the papers (122 out of 183) were conceptual, that is, not presenting an empirical base. The existing empirical papers used surveys, questionnaires, and interviews with experts, or exploited demographic and statistical data at different levels, such as at the country or company level, using various econometrical models. Furthermore, the majority (147 out of 183) discussed and/or evaluated specific potential implementations of DT initiatives and their success. Specifically, 75 papers predominantly considered positive evaluations, 50 papers argued on both positive and negative effects, whereas only 22 focused mostly on negative impacts

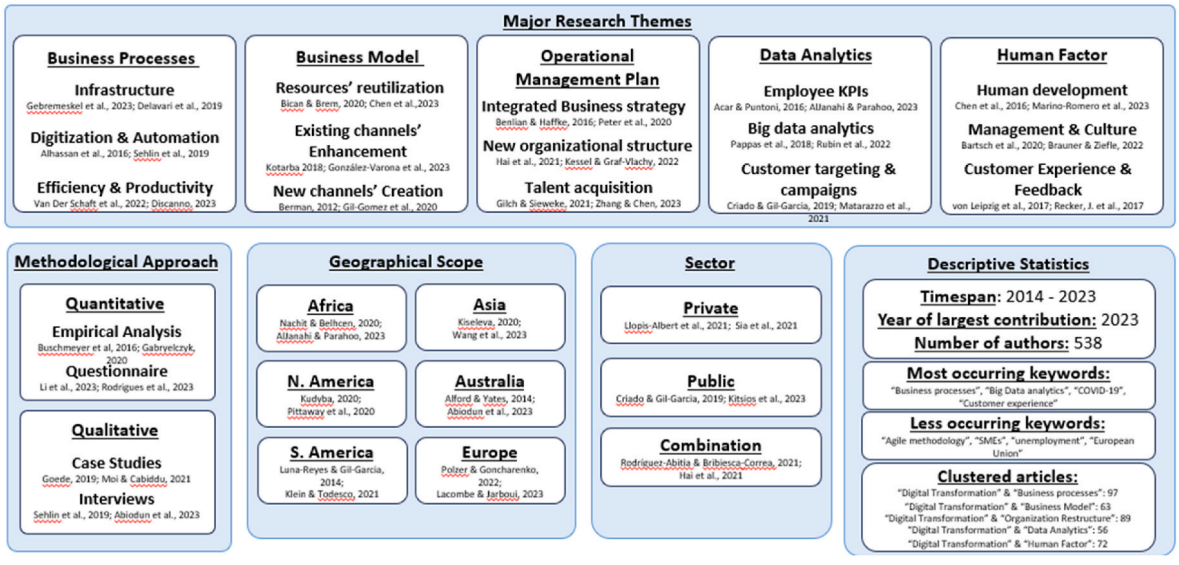


Fig. 3. Notable references regarding theoretical cornerstones, the research trends, geographical scope, methodology, and sectors.

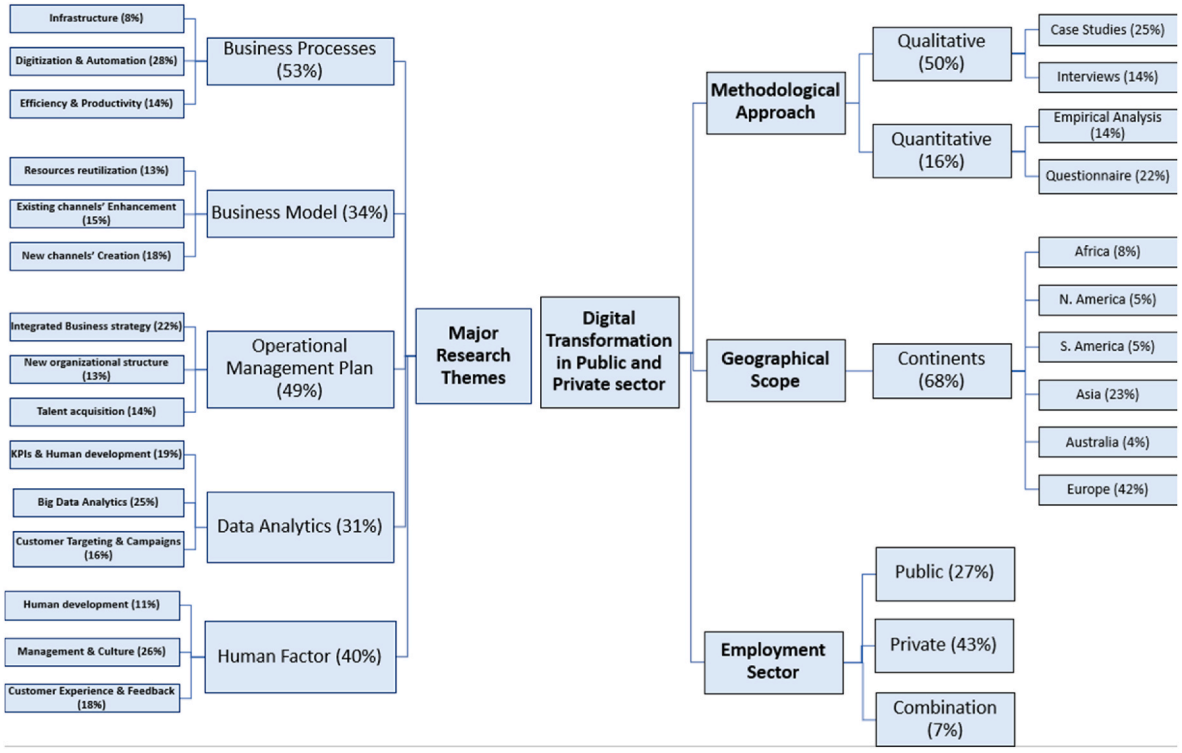


Fig. 4. Classification and characteristics of corporate factors impacted by a Digital Transformation Project.

(Fig. 4). This result reveals that the scientific viewpoint on the social opinion of digitalisation tends to be rather positive. Although the analysis was conveniently executed for the first four categories, an in-depth content review was required to extract concepts and references regarding the human factor, as most scholars presented it as an underlying attribute of the DT projects.

4.3. DTCF1: business processes digital transformation

The primary challenge most companies face relates to the extensive and bureaucratic processes that have been established internally throughout their years of operation (Shaughnessy, 2018). These processes can have a detrimental impact on both employee productivity and company efficiency (Nadkarni & Prügl, 2021).

4.3.1. DTCF1.1: infrastructure and digitisation

As suggested by the literature, prior to digitising the internal manual processes, several activities need to be conducted to prepare for the radical alterations that will be implemented. Initially, companies must update their technological and systemic infrastructure. According to Sergei et al. (2023), the removal of the old, depreciated systems and their replacement with energy-efficient, low maintenance integrated infrastructure will not only transform a company's daily operations, but also reduce its operational costs. However, as has been argued in the literature, simply updating infrastructure is insufficient, as the primary challenges employees face are the restrictions resulting from working in office premises for security and content accessibility reasons. A proven solution offered by academics is cloud computing, which has the potential to offer significant benefits to companies, such as through increasing their flexibility (e.g. working locations) and the accessibility of information remotely with minimum costs. Indeed, Buschmeyer et al. (2016) argued that advances in digital security, combined with cloud computing, can offer a safeguarded working environment, wherein unauthorised access and external threats can be heavily reduced. Following the mentioned changes, the redesign of all internal company processes is essential for them to comply with the newly applied technological model and culture (Li, Yi, Jiang, Zhang, & Zhou, 2023).

4.3.2. DTCF1.2: automation

According to scholars, the final preparatory activity should be the extensive documentation of the current operational structure, including detailed explanations of the processes and procedures (Lacombe & Jarboui, 2023; Sia et al., 2021). Applicators must first analyse the current documents, identify pain points in the execution of the processes, and thereafter, propose solutions that would offer digitisation, optimisation, and the improvement of efficiency with the removal of human involvement and manual processing where possible. Although automating processes while removing obsolete ones and replacing them with more efficient alternatives forms a considerable part of DT, it is not considered sufficient (Van Der Schaft, Lub, Van Der Heijden, & Solinger, 2022). The review revealed that many scholars have presented the digitisation of manual processes as a standalone DT project, which is almost always completed with successful results. Nonetheless, the current paper uses such digitisation as the baseline factor for the design and implementation of a robust, organisation-wide DT project (Priyono et al., 2020).

4.4. DTCF2: business model digital transformation

The next corporate factor impacted by DT projects is the re-evaluation of the corporate business model (Correani, De Massis, Frattini, Petruzzelli, & Natalicchio, 2020). Based on the literature review, a corporate business model describes the organisational scope of work intended to help a company achieve sustainability and profitability (Hai et al., 2021). Business models commonly include the products or services of a company, its target market, and its revenue streams (Matt et al., 2015). Moreover, it captures the internal model, allowing opportunities for company growth and budget management.

4.4.1. DTCF2.1: resources reutilisation

As suggested by Schallmo et al. (2017), the re-use of the workforce presents companies with an opportunity to internally transform their

business models. As the digitisation of processes can reduce the time and effort required for daily operations, employees could now be re-allocated to better support the company needs (Chen, Wang, Yao, & Lin, 2023). As the literature indicates, the increased demand for new roles and opportunities can serve to gradually benefit companies, as employees will steadily be adjusted in their new duties and operational activities. Additionally, the use of the new technological infrastructure, combined with reduced operational costs, allows companies to focus on internal departmental expansions, investment in research and development (R&D) for new products and services, and the development of staff skillsets (Vial, 2019).

4.4.2. DTCF2.2: enhancement of existing channels

Although the literature review has highlighted the tremendous competition encountered by e-shops (Imran, Shahzad, Butt, & Kantola, 2021), considering that customers could make purchases online irrespective of retailers' physical location, DT projects should focus on upgrading existing platforms to provide a modern, user-friendly, and efficient customer experience, which can subsequently increase online customer loyalty and sales. Another indicative example emerges from research on the e-commerce and logistics sectors, where companies with extensive commercial networks are heavily dependent on digital systems for their stock availability and tracking (Sutton et al., 2016). New applications of DT projects can benefit inventory operations by introducing multifunctional systems, which offer real-time assessments of stock capacity, availability, and costs, thereby significantly reducing any inconveniences to all parties involved.

4.4.3. DTCF2.3: creation of new channels

There appears to be a consensus in the literature that DT can significantly improve the development of an across-platform digital presence. Such a presence can help companies reach larger target audiences, markets, and countries. This stage involves the development of new e-shops and social media accounts for content promotion (Furjan et al., 2020). According to Discanno (2023), specific social media platforms tend to be preferred by generational groups. Therefore, organisations seeking to increase their visibility to the public and build a competitive advantage against other companies in the same sector would benefit from establishing an across-platform presence with unified content (Acar & Puntoni, 2016). The importance of having a digital presence skyrocketed during the COVID-19 pandemic, when digital channels became the only option for commercial transactions and company viability owing to governmental health restrictions and regulations. During this extremely challenging period, the EU Commission developed an emergency funding programme that fully funded the initiation of e-commence DT projects for organisations and small and medium-sized enterprises (SMEs) in all EU Member States (EU Commission, 2023). The programme was received positively by all the eligible parties, as it presented an opportunity to not only address the immediate challenge of COVID-19 restrictions, but also develop and establish a robust digital presence that would offer long term benefits (Earley, 2014; Carlsson & Rönnblom, 2022).

4.5. DTCF3: operational Management Plan digital transformation

4.5.1. DTCF3.1: integrated business strategy

DT projects are becoming increasingly integral to companies' long-term business strategies, capturing not only their immediate operational improvements, but also their strategic vision for the future. Executing a DT initiative signifies a pivotal strategic choice made by an organisation's leadership, whose decision-making process involves meticulous planning that considers numerous vital components, which prominently include allocating budgetary resources, establishing an appropriate project timeline, and the inherent complexity associated with the undertaking (Abiodun et al., 2023). Such strategic decisions are frequently incorporated into a wider organisational transformation initiative that the company is ardently dedicated to pursuing and successfully implementing (Matt et al., 2016).

4.5.2. DTCF3.2: new organisational structure

During the initial phases of the DT project, a first level analysis is conducted on potential improvements on the organisational structure and impacted areas/departments (Gurbaxani & Dunkle, 2019). DT projects reshape the organisational structure by introducing new roles, supervision processes, and operational frameworks. As digital technologies are integrated, organisations often need to establish new departments or cross-functional teams, such as digital strategy units or data governance departments, to manage the transition effectively (Furjan et al., 2020). However, these also affect the supervision process, as leadership structures begin to champion the new working culture and cultivate agility, collaboration, and efficiency among colleagues. Simultaneously, management roles should evolve to foster innovation and ensure that the newly developed digital tools will be used to enhance productivity. Upon the delivery of the DT project, the finalised structural model should depict all the organisational changes and the structural reshuffle that should be applied within the company (Erjavec et al., 2024).

4.5.3. DTCF3.3: talent acquisition

The last organisational restructure component is new-talent recruitment (Gilch & Sieweke, 2021). Young, highly trained, and well-educated candidates are primarily interested in working in modern, technologically advanced environments that offer a hybrid model (i.e. 2–3 days working from home) (Tavoletti, Kazemargi, Cerruti, Grieco, & Appolloni, 2021). Although this model received considerable criticism – with scholars exploring the risks, challenges, and negative impacts on company growth – the quarantine periods during the COVID-19 pandemic proved that companies can significantly benefit from hybrid models, as employees remain productive, while their satisfaction and appreciation towards the company increases (Schallmo et al., 2017). The recruitment of new talent is beneficial for companies as newly hired employees often introduce new working methods, modern ideas, and newly developed concepts in their areas of expertise, and are often more prompt in embracing digital culture (Chen, Wang, Yao, & Lin, 2023). Having the best talent onboard provides companies a competitive advantage, as new recruits tend to be motivated, eager to learn, and willing to take additional measures for the benefit of the company.

4.6. DTF4: data-driven digital transformation

4.6.1. DTCF4.1: KPIs & human development

It is essential for companies and organisations to have various tools available to support fair evaluations of their personnel (Berman, 2012). A DT project could support this need by developing solutions that provide reports based on set workforce key performance indicators (KPIs) (Abiodun et al., 2023). The success rate of meeting these KPIs and doing so efficiently are two measurable factors that can significantly contribute to these performance reviews (Wang, Liu, Li, & Lei, 2023). Documenting clear goals and setting a process of tracking them provides all employees the motivation to achieve them, particularly when they are linked to their career prosperity (Chatterjee, Chaudhuri, Vrontis, & Giovando, 2023). Furthermore, the corporate management is frequently updated on the performance metrics of all departments and segments via digital tools (Buschmeyer et al., 2016), which are used to evaluate the performance of each department individually, thus allowing necessary corrective actions to be identified.

4.6.2. DTCF4.2: customer targeting and campaigns

Data analytics allow companies to collect and analyse large volumes of customer data, including purchasing behaviours, browsing history, and social media interactions (Tursunbayeva et al., 2020). By using these data, businesses can create personalised recommendations, offers,

and services tailored to individual customer preferences. Consequently, companies may progress with targeted advertising campaigns based on customers' historical data and their demographic criteria, such as gender, age, and location. Scholars have emphasised different types of campaigns aimed at different audiences (Niehaves, Poeppelbuss, Plattfaut, & Becker, 2014). For instance, younger generations are driven by social media campaigns, which can be shared with either broad or highly specific audiences (Verhoef et al., 2021). However, older generations of customers appear to be more familiar with, and attracted by, more traditional means of advertising and campaigns, such as email newsletters, instant messaging, and phone campaigns. Multifunctional systems (e.g. ERP, CRM) not only provide visibility on the entire clientele database, but they also offer options for personalisation and audience targeting, allowing companies to provide company updates, information, and promotions for a better customer experience and increase in sales (Dabić et al., 2021).

4.7. DTCF5: human factor digital transformation

4.7.1. DTCF5.1: employees

Following the implementation of a DT project, a disruption in corporate business-as-usual (BAU) operations can be expected, as all end users (company management and employees) are required to adjust and become accustomed to the changes. As the human factor is considered a key contributor to the success of a DT project (Ronchi, 2022), it is vital for all employees to be trained on the new functionalities and interfaces despite their potential lack of technological knowledge and expertise. As argued by various scholars, said adjustment can prove to be somewhat challenging considering a natural disposition to change (Neumann, Winkelhaus, Grosse, & Glock, 2021), meaning that employees should be provided sufficient time to familiarise themselves with the new infrastructure (hardware) (Llopis-Albert et al., 2021) and interfaces (software) (Li, Yi, Jiang, Zhang, & Zhou, 2023). Accordingly, a change management plan needs to be designed and followed to manage these issues, helping the employees understand and embrace the new changes (Van Veldhoven & Vanthienen, 2022), and begin to use the new technological advances offered by the company. An insightful approach that appears to have successful results (according to recent studies) is the development of change ambassadors in every impacted corporate department. According to academic case studies, implementing this tactic has been found to create a positive mentality across personnel for the upcoming delivery of the DT project in question (Dabić et al., 2019). Change ambassadors are usually employees who are prone to change, eager to learn new aspects, adapt to new ideas, and enjoy working in technologically advanced working environments (Saihi et al., 2022). Thus, these employees are used as positive sponsors of DT projects, showcasing their benefits, supporting their colleagues during the transition period, and helping them adjust more rapidly and effectively. As the completion of a DT project includes (as one of its most critical aspects) the acceptance of its deliverables by the employees (Kane, 2019), failure of acceptance will lead to the rejection of the new tools and processes, causing extensive disturbance to the company's Business as Usual (BAU) operations (Carlsson & Rönnblom, 2022). This disruption manifests in various forms and degrees of impact, beginning with dissatisfaction and potentially culminating in employee resignations. This has primarily been observed in cases where the employees are unable to adjust to the changes of a newly digitised working environment (Sergei et al., 2023; Palagan, 2025).

4.7.2. DTCF5.2: executive leadership

Recent case studies have demonstrated a radical cultural shift within companies following the application of a DT project, where company administrations have transitioned to more agile, collaborative structures, steadily diverging from the traditional hierarchy of silos (Nadkarni & Prügl, 2021). Frequently, the transition period, post the application of the DT project, is somewhat challenging for the personnel involved as they attempt to adjust to their new working reality. During this period, team leaders and executive leadership teams are required to be active and engaging, guiding their workforce throughout the transition period (Li, Yi, Jiang, Zhang, & Zhou, 2023). The literature has extensively documented that introducing a digital working culture drives a significant increase in employee satisfaction (Saunavaara et al., 2022; Sehlin et al., 2019). This cultural shift, driven by team leaders across departments, provides the employees the space to focus on their professional development, and expand their talents and contributions to the company's growth. Consequently, employee productivity and morale are enhanced, contributing to the development of a positive and stable work environment (Gurbaxani & Dunkle, 2019).

Although executive leadership is expected to drive the change, scholars have highlighted that resistance towards the digital initiative can also be found among management members. Senior employees can also feel threatened by radical transformations as they may lose their authority or teams' responsibility (Discanno, 2023). As the literature suggests, these voices of disagreement may represent the minority of cases in changes to management processes. However, to foster the long-term success of the initiative, their adjustment must be handled while the core management team proceeds with the transformation (Acar & Puntoni, 2016).

4.7.3. DTCF5.3: customer experience and satisfaction

Companies frequently identify gaps in customer services or reduction in customer purchases owing to unpleasant or unsatisfactory experiences (Hindle, Kunc, Mortensen, Oztekin, & Vidgen, 2020). DT projects can provide solutions to most of the problems related to the customer experience as all customer transactions can become more streamlined, efficient, and modernised (Gurbaxani & Dunkle, 2019). A major factor that positively impacts the customer experience is the increased service speed. The development of streamlined internal processes and procedures (DTCF1) allows employees to execute customer enquiries and orders in a timely, delay-free manner. The automation of flows drives a reduction in interactions among customers and employees, subsequently reducing human errors or mishandling (Schwertner, 2017). Moreover, the customers are provided the time and space necessary to execute any manual actions as per their convenience, without feeling pressurised (Dabić et al., 2021; Shaughnessy, 2018). Through these options, as well as with the removal of anxieties over not having visibility or tracking, customers can feel a greater sense of control over their purchase or enquiry (Ren et al., 2023).

5. Discussion

The thematic and content analyses of the academic literature offer several conclusions regarding the research questions. The first research question, concerning the four traditional categories of corporate components, has been extensively analysed in the existing literature, with the term 'business processes' featuring in most of the reviewed articles. Several studies analysed more than one corporate factor, identifying or producing new connections and interdependencies among them. Following the extensive combination of two or more categories, a pattern can be formed indicating that an application of a DT project should be considered successful only if it has effectively altered all the traditional corporate factors simultaneously.

The second research question reviewed whether the human factor could be introduced as the fifth essential factor for the successful application of a DT project. The initial search in databases yielded limited references with human factor terms (e.g. human factor, employee, management, customer) in titles or abstracts. Contrarily, the second-level analysis (content analysis) identified that for many scholars the human factor may not be treated as an individual aspect, but it is always included as a facilitator or moderator in the digitisation of the traditional corporate factors (e.g. business processes, employees). Such topics as people management, job automation, skilling and re-skilling,

management change, and digital customer experience have regularly appeared in many articles and manuscripts. To highlight the importance of the human factor in the success of DT projects, several studies analysed failed DT case studies. These cited the absence of consultation and collaboration with the people impacted by the DT project (in any of the DT implementation phases) as one of the most important failure factors.

5.1. Implications

5.1.1. Theoretical implications

As identified in the literature, the human factor plays a significant role in the successful application of a DT project. However, to date, a few studies have targeted this factor as a crucial denominator for the success of a DT project. Doukidis and Spinellis (2020) indicatively presented a quad-pillar model including all the key corporate factors (and their subcategories) that are impacted during a DT project. Most of the academic studies contributed to the analysis of these factors either partially or holistically, using the human factor as a facilitator to the occurring transformational changes rather than as a distinct category in and of itself.

Following our extensive research, we propose an updated, enhanced penta-factor model, including a new pillar, the human factor, divided into three sub-categories: corporate personnel, corporate management, and customers. The sub-category 'corporate personnel' comprises all the initiatives to be applied to employees during a DT project, which must be accepted and embraced in the new digital working reality (Loonam, Eaves, Kumar, & Parry, 2018). As humans are prone to routine and constants, it is highly important for DT project teams to promote all the benefits and improvements the employees will have post the implementation of the project to ease their adjustment.

The second sub-category refers to company management, who represent the initiators of this radical organisational change. The management could take actions to champion this change and foster a digital culture across the company (Hashem et al., 2015). It is highly important to include the opportunity offered to the management for organisational restructure based on the transformed working model. New departments could be created supporting the digital functions of the company, employees can be re-used and transferred among departments to further develop their skillset and talents, and new talent with digital skills could be recruited to support the company on this transition (Kane et al., 2019). Following the restructure and the subsequent reduction of operational costs, the company management could offer various reporting tools regarding employee and customer satisfaction, thereby helping create an environment where the company's viability and profitability can flourish.

The final proposed sub-category is the customers. Although extensive analysis has already been conducted in customer experience, particularly within e-commerce, we introduce the term as part of the sociological concept of a human factor component accepting and embracing the change. DT projects which offer a unified expanded digital presence, target broad audiences, extended markets, and multiple generations of people (Niehaves, Poeppelbuss, Plattfaut, & Becker, 2014). Therefore, this digital presence must consider all the generational limitations (e.g. limited technological experience) and people with disabilities who face difficulties when accessing e-shops and e-commerce platforms. A recent example of this was the e-banking model in Greece during the COVID-19 pandemic. A related study (Vratimos, 2023) highlighted that the National Bank of Greece (NBG) rose to the occasion to support its customers by creating videos with clear instructions, a step-by-step visualisation of the process, and a well-known actress to present the initiative.

Moreover, this sub-category includes the concept of 'customer loyalty', which is tremendously significant to the customers themselves. Extensive literature has demonstrated that customers select specific retailers for their purchases when they feel appreciated, respected, and have a tailored experience. All these attributes can now be effectively served through the deliverables of a DT project. Loyalty schemes, personalised campaigns, and customer rewards are but some of the methods for improving customer satisfaction.

Accordingly, the proposed penta-factor model includes the following DT pillars: a) business processes, b) business model, c) organisational management, d) data-driven digital transformation, and e) human factor (see Fig. 5).

5.1.2. Practical implications

As discussed above, to increase successful application rates, future applicators could develop frameworks that focus on the human factor across all stages of the DT project to safeguard its trajectory. The framework that has been produced as the outcome of the current analysis follows the penta-factor model, as it includes specific actions and tasks for the parties involved, and highlights the crucial role of human factor in all major transformative activities. The framework comprises nine action points (ACs) that could be expanded in greater detail, depending on the project's scope, size, and complexity.

AC1: Produce an in-depth analysis for the current operational structure of the organisation and extensive documentation on all the modifications that will be applied to inform all impacted parties.

AC2: Develop a change in management strategy to tackle and accommodate the resistance that will be met in all workforce layers.

AC3: Different categories and levels of employees could be consulted during every stage of a DT project's development. These employees' input regarding specific modifications or project deliverables can enhance the project's scope and lead to its long-term implementation.

AC4: Introduce change ambassador teams during the last stages of the project across the company layers to support employees' acceptance and adoption of the new working methods.

AC5: Management teams who have selected DT projects as part of their corporate strategy plan could actively embrace the digital-first culture when engaging in activities with their employees. Developing a culture of agility, collaboration, and cooperation across departments will allow the employees to steadily adapt to their new digital working environment.

AC6: Training is an integral part of the DT project for all end users, that is, employees and upper management using the system from the company side, and the customers who will use the new platforms and digital means of communication for their transactions.

AC7: Human Resources departments must target the recruitment of young, technologically engaged employees to help companies enhance their personnel and increase the number of employees who prefer digital working environments and state-of-the-art equipment.

AC8: Training models could also be developed for customers when new features are first introduced. Customers who are used to specific user interface or user experience models may find adjustments to be challenging or alienating. Therefore, engaging methods should be developed to onboard the existing clientele towards the revamped digital channels while introducing the new digital presence to new markets or customers.

AC9: Finally, applicators would do well to highlight all the potential ways that the DT project will increase customer experience and satisfaction, particularly as the viability, sustainability, and profitability of a company is based on customer volumes and repeat purchases. Increasing the customer satisfaction will lead to increased sales and company growth.

6. Conclusion and future research

This study offers a holistic approach to all the factors (which will be heavily impacted and altered) that need to be considered and agreed upon by the corporate management. Using a robust systematic cluster analysis, the outcomes of this study have been generated by a wide range of academic concepts and provide valuable insights for future scholars. This study has attempted to create a conceptual framework that includes all the success criteria to be considered when evaluating future DT

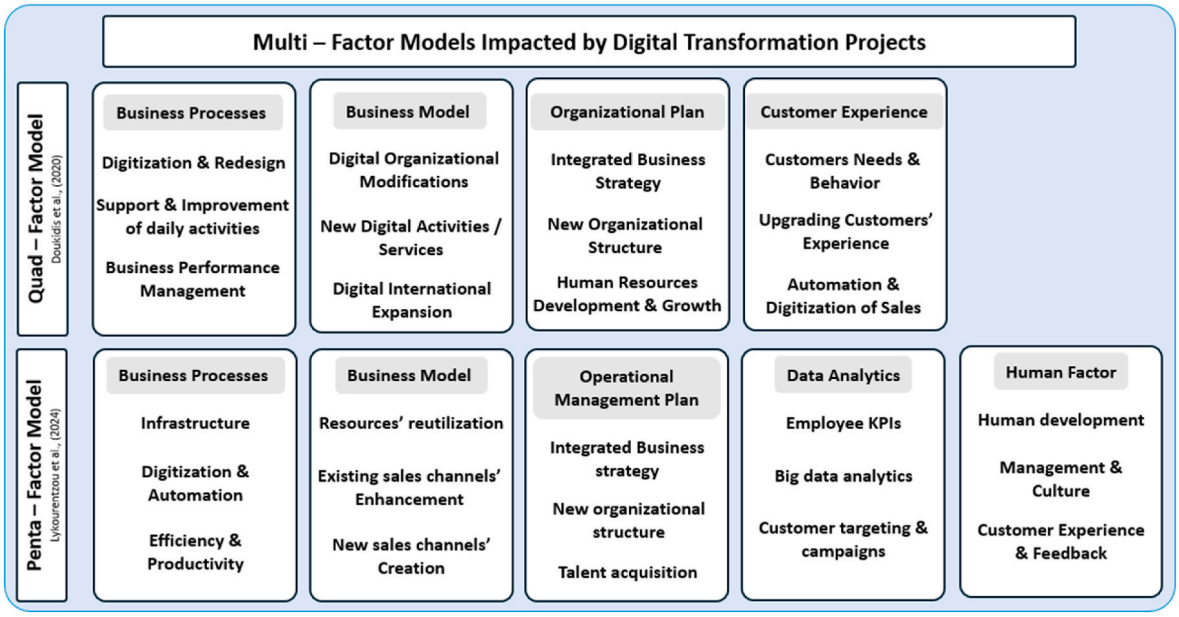


Fig. 5. The quad - factor model by Doukidis et al. (2020) and the penta - factor model by Lykourentzou et al., (2024).

implementations, as well as a summary of the identified implications that are worthy of further exploration. DT projects are in a continuous demand by organisations in the public and private sectors. As such, new case studies and data are being consistently produced, meaning that further analyses and academic research are well-positioned to improve the rates of successful application.

However, this study is not without limitations. First, articles and publications from conferences and grey literature were excluded as they are considered less robust than journal articles from a scientific perspective. Second, a limitation commonly associated with literature reviews is the inherent subjectivity involved in selecting and analysing the articles. For example, the choice of keywords for identifying relevant documents and conducting the bibliographic analysis, although informed by previous studies, may have influenced the results. Finally, another potential limitation is the tendency of certain methodologies to prioritise evaluating the most frequently discussed topics. Consequently, this may have resulted in excluding the less frequently debated, yet practically significant, aspects from our analysis. To address these issues and support the scholars interested in a more in-depth discussion of the specific concepts identified in this literature review, further research needs to be conducted to provide a more extensive documentation. Thus, additional outcomes can be identified, which could serve to enhance the penta-factor model presented in this literature review.

Moreover, extensive research is required to more accurately capture and analyse how all the groups involved in the transformation (e. g. employees, company administration, customers) will be impacted by, and absorb, the change.

Furthermore, most of the case studies have been conducted in large companies and international organisations with a high number of employees, complex procedures and processes, predefined budgets, and established business models guided by firm organisational structures. Consequently, additional research on SMEs that implement DT projects would help to expand the existing literature. Using its cohesion policy and several funding programmes, the European Union (EU) has promoted funding projects for the digital establishment and modernisation of SMEs. As these projects are ongoing, they could become the subject of future research and analysis focusing on the successful implementation of projects under challenging timeframes. This study could identify the policy framework established by the EU for the continuous support of SMEs through funding programmes. Further conclusions could emerge regarding the viability of the funded SMEs and the link to this viability with EU funding.

In addition, during our search for related articles and papers, the most relevant literature either analysed or showcased DT projects applied in the private sector. It would be equally important for extensive research to be developed on cases of public-sector DT projects. Topics that can further be discussed and analysed include the reasons underlying the decisions to implement DT projects within the public sector, as well as how these applications could increase citizens' satisfaction and trust towards the state.

Finally, most studies have focused on the beneficial attributes of a DT project. This is owing to how the benefits are easily noticeable, which, in turn, is why an increasing number of companies are proceeding with DT projects. Nevertheless, capturing the challenges that DT projects face, including their mitigation actions, is equally essential.

CRediT authorship contribution statement

Maria Anna Lykourentzou: Writing – original draft, Investigation, Formal analysis, Data curation. **Nikolaos Apostolopoulos:** Writing – review & editing, Writing – original draft, Validation. **Marina Dabić:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Panagiotis Liargovas:** Writing – original draft, Validation. **Metka Tekavčič:** Writing – review & editing, Writing – original draft, Funding acquisition.

Acknowledgments

This research was financially supported by the Slovenian Research Agency (www.arrs.gov.si) within the research program P5–0441. The funders had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.techsoc.2025.102934.

Data availability

Data will be made available on request.

References

Abiodun, T. S., Rampersad, G., & Brinkworth, R. (2023). Driving smartness for organizational performance through industry 4.0: A systems perspective. *Journal of Manufacturing Technology Management, 34*(9), 40–63 [CrossRef].

Acar, O. A., & Puntoni, S. (2016). Customer empowerment in the digital age. *Journal of Advertising Research, 56*(1), 4–8 [CrossRef].

Berman, S. J. (2012). Digital transformation: Opportunities to create new business models. *Strategy & Leadership, 40*(2), 16–24 [CrossRef].

Buschmeyer, A., Schuh, G., & Wentzel, D. (2016). Organizational transformation towards product-service systems – empirical evidence in managing the behavioral transformation process. *Procedia CIRP, 47*, 264–269 [CrossRef].

Carlsson, V., & Rönnblom, M. (2022). From politics to ethics: Transformations in EU policies on digital technology. *Technology in Society, 71*(102145), Article 102145 [CrossRef].

Chatterjee, S., Chaudhuri, R., Vrontis, D., & Giovando, G. (2023). Digital workplace and organization performance: Moderating role of digital leadership capability. *Journal of innovation & knowledge, 8*(1), Article 100334 [CrossRef].

Chen, C., Wang, S., Yao, S., & Lin, Y. (2023). Does digital transformation increase the labor income share? From a perspective of resources reallocation. *Economic Modelling, 128*(106474), Article 106474 [CrossRef].

Correani, A., De Massis, A., Frattini, F., Petruzzelli, A. M., & Natalicchio, A. (2020). Implementing a digital strategy: Learning from the experience of three digital transformation projects. *California Management Review, 62*(4), 37–56 [CrossRef].

Dabić, M., Lažnjak, J., Smallbone, D., & Švarc, J. (2019). Intellectual capital, organisational climate, innovation culture, and SME performance: Evidence from Croatia. *Journal of Small Business and Enterprise Development, 26*(4), 522–544 [CrossRef].

Dabić, M., Marzi, G., Vlačić, B., Daim, T. U., & Vanhaverbeke, W. (2021). 40 years of excellence: An overview of Technovation and a roadmap for future research. *Technovation, 106*(102303), Article 102303 [CrossRef].

Datta, P., & Nwankpa, J. K. (2021). Digital transformation and the COVID-19 crisis continuity planning. *Journal of Information Technology Teaching Cases, 11*(2), 81–89 [CrossRef].

Discanno, F. (2023). A comparison between the labor market trends in the European and Italian banking sectors: The impact of digital transformation and the related need for investing in the human factor. *Cambridge Open Engage*. https://doi.org/10.33774/coe-2023-1cldg

Doukidis, G., Spinellis, D., & Ebert, C. (2020). Digital transformation - a primer for practitioners. *IEEE software, 37*(5), 13–21 [CrossRef].

Earley, S. (2014). The digital transformation: Staying competitive. *IT professional, 16*(2), 58–60 [CrossRef].

Erjavec, J., Indihar Štemberger, M., & Jaklič, J. (2024). How to develop organizational forms for a successful digital transformation? Findings from two case studies. *Journal of the knowledge economy, 15*, 9576–9596. https://doi.org/10.1007/s13132-023-01479-3.

Furjan, M. T., Tomičić-Pupek, K., & Pihir, I. (2020). Understanding digital transformation initiatives: Case studies analysis. *Business Systems Research Journal, 11*(1), 125–141 [CrossRef].

Gaß, O., Ortbach, K., Kretzer, M., Maedche, A., & Niehaves, B. (2015). Conceptualizing individualization in information systems – a literature review. *Communications of the Association for Information Systems, 37* [CrossRef].

Gilch, P. M., & Sieweke, J. (2021). Recruiting digital talent: The strategic role of recruitment in organisations' digital transformation. *German journal of human resource management, 35*(1), 53–82 [CrossRef].

Grybauskas, A., Stefanini, A., & Ghobakhloo, M. (2022). Social sustainability in the age of digitalization: A systematic literature review on the social implications of industry 4.0. *Technology in Society, 70*(101997), Article 101997 [CrossRef].

Gurbaxani, V., & Dunkle, D. (2019). Gearing up for successful digital transformation. *MIS Quarterly Executive, 18*(3), 209–220 [CrossRef].

Hai, T. N., Van, Q. N., & Thi Tuyet, M. N. (2021). Digital transformation: Opportunities and challenges for leaders in the emerging countries in response to Covid-19 pandemic. *Emerging science journal, 5*, 21–36 [CrossRef].

Hashem, I. A. T., Yaqoob, I., Anuar, N. B., Mokhtar, S, Gani, A., & Khan, S. U (2015). The rise of 'big data' on cloud computing: Review and open research issues. *Information Systems, 47*, 98–115 [CrossRef].

Higgins, J. P. T. (2019). In J. Higgins, & J. Thomas (Eds.), *Cochrane handbook for systematic reviews of interventions* (2nd ed.). Hoboken, NJ: Wiley-Blackwell [CrossRef].

Hindle, G., Kunc, M., Mortensen, M., Oztekin, A., & Vidgen, R. (2020). Business analytics: Defining the field and identifying a research agenda. *European Journal of Operational Research, 281*(3), 483–490 [CrossRef].

Imran, F., Shahzad, K., Butt, A., & Kantola, J. (2021). Digital transformation of industrial organizations: Toward an integrated framework. *Journal of Change Management, 21*(4), 451–479 [CrossRef].

Ivančić, L., Vukšić, V., & Spremić, M. (2019). Mastering the digital transformation process: Business practices and lessons learned. *Technology Innovation Management Review, 9*(2), 36–50 [CrossRef].

Kane, G. (2019). The technology fallacy: People are the real key to digital transformation. *Research-Technology Management, 62*(6), 44–49 [CrossRef].

Kempeneer, S., & Heylen, F. (2023). Virtual state, where are you? A literature review, framework, and agenda for failed digital transformation. *Big Data & Society, 10*(1), Article 205395172311605 [CrossRef].

Kitchenham, B. (2004). *Procedures for performing systematic reviews*. Keele, UK: Keele Univ. Keele, UK, Keele Univ.

Kraus, S., Jones, P., Kailer, N., Weinmann, A., Chaparro-Banegas, N., & Roig-Tierno, N. (2021). Digital transformation: An overview of the current state of the art of research. *Sage Open, 11*(3), Article 215824402110475 [CrossRef].

Kudyba, S. (2020). COVID-19 and the acceleration of digital transformation and the future of work. *Information Systems Management, 37*(4), 284–287 [CrossRef].

Lacombe, I., & Jarboui, A. (2023). Governance and management of digital transformation projects: An exploratory approach in the financial sector. *International Journal of Innovation Science, 15*(4), 611–635 [CrossRef].

Li, F. (2020). Leading digital transformation: Three emerging approaches for managing the transition. *International Journal of Operations & Production Management, 40*(6), 809–817 [CrossRef].

Li, L., Yi, Z., Jiang, F., Zhang, S., & Zhou, J. (2023). Exploring the mechanism of digital transformation empowering green innovation in construction enterprises. *Developments in the Built Environment, 15*(100199), Article 100199 [CrossRef].

Llopis-Albert, C., Rubio, F., & Valero, F. (2021). Impact of digital transformation on the automotive industry. *Technological Forecasting and Social Change, 162*(120343), Article 120343 [CrossRef].

Loonam, J., Eaves, S., Kumar, V., & Parry, G. (2018). Towards digital transformation: Lessons learned from traditional organizations. *Strategic Change, 27*(2), 101–109 [CrossRef].

Matt, C., Hess, T., & Benlian, A. (2015). Digital transformation strategies. *Business & Information Systems Engineering, 57*(5), 339–343 [CrossRef].

Mergel, I., Edelmann, N., & Haug, N. (2019). Defining digital transformation: Results from expert interviews. *Government Information Quarterly, 36*(4), Article 101385 [CrossRef].

Nadkarni, S., & Prügl, R. (2021). Digital transformation: A review, synthesis and opportunities for future research. *Management Review Quarterly, 71*(2), 233–341 [CrossRef].

Neumann, W. P., Winkelhaus, S., Grosse, E. H., & Glock, C. H. (2021). Industry 4.0 and the human factor – a systems framework and analysis methodology for successful development. *International Journal of Production Economics, 233*(107992), Article 107992 [CrossRef].

Niehaves, B., Poeppelbuss, J., Plattfaut, R., & Becker, J. (2014). BPM capability development – a matter of contingencies. *Business Process Management Journal, 20*(1), 90–106 [CrossRef].

Okoli, C., & Schabram, K. (2010). A guide to conducting a systematic literature review of information systems research. *Sprouts: Working Papers on Information Systems, 10*(26). http://sprouts.aisnet.org/10-26.

Pappas, I. O., Mikalef, P., Giannakos, M. N., Krogstie, J., & Lekakos, G. (2018). Big data and business analytics ecosystems: Paving the way towards digital transformation and sustainable societies. *Information Systems and e-Business Management, 16*(3), 479–491 [CrossRef].

Petticrew, M., & Roberts, H. (2006). *Systematic reviews in the social sciences: A practical guide*. Blackwell Publishing.

Priyono, A., Moin, A., & Putri, V. N. A. O. (2020). Identifying digital transformation paths in the business model of SMEs during the COVID-19 pandemic. *Journal of Open Innovation Technology Market and Complexity, 6*(4), 104 [CrossRef].

Ren, T., Liu, X., & Ding, J. (2023). Intergenerational dynamics of digital transformation in family firms. *Technology in Society, 74*(102261), Article 102261 [CrossRef].

Ronchi, A. M. (2022). Human factor, resilience, and cyber/hybrid influence. *Information and security : an international journal, 53*, 221–239 [CrossRef].

Saihi, A., Ben-Daya, M., & As'ad, R. (2022). A survey of the underlying success factors of maintenance digital transformation. *IFAC-PapersOnLine, 55*(10), 2944–2949 [CrossRef].

Saunavaara, J., Laine, A., & Salo, M. (2022). The Nordic societies and the development of the data centre industry: Digital transformation meets infrastructural and industrial inheritance. *Technology in Society, 69*(101931), Article 101931 [CrossRef].

Schallmo, D., Williams, C. A., & Boardman, L. (2017). Digital transformation of business models — best practice, enablers, and roadmap. *International Journal of Innovation Management, 21*(8), Article 1740014 [CrossRef].

Schwertner, K. (2017). Digital transformation of business. *Trakia Journal of Science, 15*(Suppl.1), 388–393 [CrossRef].

Sehlin, D., Truedsson, M., & Cronemyr, P. (2019). A conceptual cooperative model designed for processes, digitalisation and innovation. *International Journal of Quality and Service Sciences, 11*(4), 504–522 [CrossRef].

Sergei, T., Arkady, T., Natalya, L., Pathak, R. D., Samson, D., Husain, Z., & Sushil, S. (2023). Digital transformation enablers in high-tech and low-tech companies: A comparative analysis. *Australian Journal of Management, 48*(4), 801–843 [CrossRef].

Shaughnessy, H. (2018). Creating digital transformation: Strategies and steps. *Strategy & Leadership, 46*(2), 19–25 [CrossRef].

Sia, S. K., Weill, P., & Zhang, N. (2021). Designing a future-ready enterprise: The digital transformation of DBS Bank. *California Management Review, 63*(3), 35–57 [CrossRef].

Siddaway, A. P., Wood, A. M., & Hedges, L. V. (2019). How to do a systematic review: A best practice guide for conducting and reporting narrative reviews, meta-analyses, and meta-syntheses. Meta-analyses. *Annual Review of Psychology, 70*(1), 747–770.

Spaltini, M., Acerbi, F., Pinzone, M., Gusmeroli, S., & Taisch, M. (2022). Defining the roadmap towards industry 4.0: The 6Ps maturity model for manufacturing SMEs. *Procedia CIRP, 105*, 631–636 [CrossRef].

Strohmeier, S. (2020). Digital human resource management: A conceptual clarification. *German journal of human resource management, 34*(3), 345–365 [CrossRef].

Sutton, A., Papaioannou, D., & Booth, A. (2016). In *Systematic approaches to a successful literature review* (2nd ed.). London, England: SAGE Publications.

Tavoletti, E., Kazemargi, N., Cerruti, C., Grieco, C., & Appolloni, A (2021). Business model innovation and digital transformation in global management consulting firms. *European Journal of Innovation Management, 25*(6), 612–636 [CrossRef].

Teichert, R. (2019). Digital transformation maturity: A systematic review of literature. *Acta Universitatis Agriculturae et Silviculturae Mendelianae Brunensis, 67*(6), 1673–1687 [CrossRef].

Tranfield, D., Denyer, D., & Smart, P. (2003). Towards a methodology for developing evidence-informed management knowledge by means of systematic review. *British Journal of Management, 14*(3), 207–222 [CrossRef].

Tursunbayeva, A., Bunduchi, R., & Pagliari, C. (2020). Planned benefits' can be misleading in digital transformation projects: Insights from a case study of human resource information systems implementation in healthcare. *Sage Open, 10*(2), Article 215824402093388 [CrossRef].

Van Der Schaft, A. H. T., Lub, X. D., Van Der Heijden, B., & Solinger, O. N. (2022). How employees experience digital transformation: A dynamic and multi-layered sensemaking perspective. *Journal of Hospitality & Tourism Research*, Article 109634802211230 [CrossRef].

Van Veldhoven, Z., & Vanthienen, J. (2022). Digital transformation as an interaction-driven perspective between business, society, and technology. *Electronic Markets, 32*(2), 629–644 [CrossRef].

Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Dong, J. Q., Fabian, N., & Haenlein, M. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research, 122*, 889–901 [CrossRef].

Vial, G. (2019). Understanding digital transformation: A review and a research agenda. *The Journal of Strategic Information Systems, 28*(2), 118–144 [CrossRef].

Vratimos, A. (2023). The Emergence of the Hybrid Working Model During COVID-19: The Case of Greece. In *Corporate Growth and Organisational Development – jobily.gr* (accessed April 2024) https://jobily.gr/blog/6b286d77-024a-4653-accb-cc571813791c.

Wang, X., Liu, Z., Li, J., & Lei, X. (2023). How organizational unlearning leverages digital process innovation to improve performance: The moderating effects of smart technologies and environmental turbulence. *Technology in Society, 75*(102395), Article 102395 [CrossRef].

Williams, C., & van Triest, S. (2023). Understanding performance in professional services for innovation intermediation: Technology consultants vs. management consultants. *Technovation, 126*(102824), Article 102824 [CrossRef].

Zaki, M. (2019). Digital transformation: Harnessing digital technologies for the next generation of services. *Journal of Services Marketing, 33*(4), 429–435 [CrossRef].

Zaoui, F., & Souissi, N. (2020). Roadmap for digital transformation: A literature review. *Procedia Computer Science, 175*, 621–628 [CrossRef].

Zhang, J., & Chen, Z. (2024). Exploring human resource management digital transformation in the digital age. *Journal of the knowledge economy, 15*, 1482–1498. https://doi.org/10.1007/s13132-023.

Zhu, X., Ge, S., & Wang, N. (2021). Digital transformation: A systematic literature review. *Computers & Industrial Engineering, 162*(107774), Article 107774 [CrossRef].